Stretching the Manifold: The Ambient-Intrinsic Dimension Gap as a Universal Predictor of Machine Learning Algorithm Behavior

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Abstract

Algorithms built for low-dimensional data are routinely applied to data with hundreds or thousands of features, often with unpredictable results: some fail outright, others barely notice. Existing explanations for this focus on either the ambient dimension of the data or its intrinsic dimension, but neither on its own reliably predicts which outcome to expect. We argue that what matters is the *ratio* between the two. We call this the stretching ratio, $\boldsymbol{\rho = d_a/d_i}$, and show across 45 controlled experiments that it predicts $\boldsymbol{k}$-means clustering failure remarkably well: performance follows a power law in $\boldsymbol{\rho}$, with a sharp transition around $\boldsymbol{\rho^* \approx 3.5}$–$\mathbf{5}$ that holds regardless of the underlying intrinsic dimension. The decay rate itself depends on $\boldsymbol{d_i}$ in a simple way, $\boldsymbol{\beta(d_i) = 0.140\ln(d_i) + 0.534}$, accurate to within 0.025 across all conditions tested. More surprisingly, $\boldsymbol{k}$NN-based classification and regression move in the *opposite* direction – both improve as $\boldsymbol{\rho}$ increases. This is a consistent, statistically significant trend, and it points to a genuine geometric tradeoff rather than a single curse of dimensionality. We also find that $\boldsymbol{\log(\rho)}$ alone explains more variance in clustering performance ($\boldsymbol{R^2 = 0.556}$) than participation ratio, explained variance ratio, or ambient/intrinsic dimension individually, and that heavy-tailed noise pushes the failure threshold even lower, below $\boldsymbol{\rho^* = 2}$. Taken together, these results give practitioners something concrete: estimate $\boldsymbol{\rho}$ before clustering, and treat anything above 5 as a warning sign.

Keywords: Curse of dimensionality, Intrinsic dimension, Stretching ratio, k-means clustering, Manifold hypothesis, Scaling laws

1 Introduction

Anyone who has run k-means on a dataset with a few hundred features has probably seen it fail in a way that's hard to explain. The same algorithm might work perfectly on a 10-dimensional version of the same data. The *curse of dimensionality* is the standard name for this phenomenon, and it has been studied for decades, yet there is still no simple, computable rule that tells a practitioner *in advance* whether their specific dataset and algorithm combination is going to work.

Two lines of explanation dominate the literature. One looks at *ambient dimension* d_a – simply, how many features the data has. The other, motivated by the manifold hypothesis, looks at *intrinsic dimension* d_i – the actual number of degrees of freedom underlying the data, which is often far smaller than d_a. Both ideas capture something real, but neither one alone turns out to be a good predictor of failure across different settings. A dataset with $d_a = 1000$ and $d_i = 5$ behaves very differently from one with $d_a = 10$ and $d_i = 5$, even though d_i is identical in both cases.

This observation is the starting point for our work. What if the relevant quantity isn't d_a or d_i on its own, but how far apart they are – how much the data manifold is "stretched" inside its ambient space? We define the *stretching ratio*:

$$\rho = \frac{d_a}{d_i} \tag{1}$$

At $\rho = 1$, the data fills its ambient space efficiently – there's no wasted room. As ρ grows, more and more ambient dimensions carry nothing but noise, and that noise dilutes the pairwise distances that many algorithms rely on.

Contributions.

Concretely, this paper makes the following contributions:

- We show that k-means clustering failure follows a power law in ρ: $\mathrm{ARI}(\rho) \approx \alpha \cdot \rho^{-\beta}$, with a phase transition at $\rho^* \approx 3.5$–5 that appears consistently across different intrinsic dimensions.
- We find that the decay exponent β itself follows a simple logarithmic law in d_i: $\beta(d_i) = 0.140 \ln(d_i) + 0.534$, with prediction error under 0.025.
- We show that kNN-based methods move in the *opposite* direction – improving as ρ increases – which points to an algorithm-dependent geometric tradeoff rather than a uniform "curse."
- We compare $\log(\rho)$ against several existing complexity measures (participation ratio, explained variance ratio) and find it is the stronger predictor of clustering failure ($R^2 = 0.556$ vs. 0.321).
- We show that heavy-tailed noise shifts ρ^* downward, connecting our framework to known distributional failure modes.

2 Related Work

Curse of dimensionality.

The basic observation that distance-based methods degrade in high dimensions goes back at least to Bellman [1]. Beyer et al. [2] later sharpened this considerably, showing that under fairly general conditions, the very notion of a "nearest neighbor" loses meaning as dimension grows – the distances to the nearest and farthest points converge. A number of follow-up studies have characterized exactly how fast this convergence happens for different distributions.

Intrinsic dimensionality.

A separate body of work starts from the manifold hypothesis [3]: the idea that real-world high-dimensional data actually lives on (or near) a much lower-dimensional manifold. This raises the practical question of how to estimate that lower dimension, and several estimators have been proposed, including the TwoNN method [4] and maximum likelihood approaches [5]. We use TwoNN throughout this paper, mainly because it requires no tuning and works well on the modest sample sizes in our experiments.

Distance concentration and scaling laws.

More recent work has tried to pin down empirical scaling laws for how distance contrast degrades with dimensionality, and has found that these laws hold fairly consistently for distributions with finite third moment. Our work sits in a similar spirit but asks a different question: rather than treating ambient dimension as the sole driver of degradation, we treat the *ratio* between ambient and intrinsic dimension as the governing quantity, and we look at how this plays out not just for clustering but for classification and regression as well.

Adversarial vulnerability and the dimension gap.

The gap between ambient and intrinsic dimension has also come up in work on adversarial robustness, where a larger gap has been linked to greater vulnerability to adversarial perturbations. Our results can be read as a generalization of that observation – the same gap appears to matter for "ordinary" algorithm failure too, not just adversarial settings.

3 Theoretical Framework

Definition 1 (Stretching Ratio) *Let $d_a \in \mathbb{N}$ denote the ambient dimension of a dataset $\mathbf{X} \in \mathbb{R}^{n \times d_a}$, and let d_i denote its intrinsic dimension as estimated by TwoNN. We define the stretching ratio as*

$$\rho = \frac{d_a}{d_i}, \quad \rho \geq 1. \tag{2}$$

Definition 2 (Stretching Gap) *The stretching gap is $\delta = d_a - d_i$, the absolute number of "extra" dimensions that carry no signal.*

Why should this matter?

Think of the d_i dimensions as carrying the actual structure of the data – the part that distinguishes one cluster from another – and the remaining $d_a - d_i$ dimensions as carrying nothing but noise. If that noise is roughly independent across dimensions, each additional noise dimension adds a small amount of variance to every pairwise distance, regardless of which cluster the points belong to. The more noise dimensions there are relative to signal dimensions, the more this added variance swamps the actual signal. In rough terms, the signal-to-noise ratio in pairwise distances should scale like $d_i/d_a = 1/\rho$.

We can make this slightly more precise.

Proposition 1 (Distance Contrast Degradation) *Let $\mathbf{X} \in \mathbb{R}^{n\times d_a}$ be a dataset whose signal lies on a d_i-dimensional subspace, with isotropic Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2\mathbf{I}_{d_a})$ added in all ambient dimensions. Define the distance contrast ratio as $\Delta(\rho) = \mathbb{E}[\|\mathbf{x}_i - \mathbf{x}_j\|]/\mathrm{std}[\|\mathbf{x}_i - \mathbf{x}_j\|]$, where the expectation is taken over pairs of points drawn from different clusters. Then $\Delta(\rho)$ degrades as*

$$\Delta(\rho) \;=\; \Omega\Big(\rho^{-1/2}\Big)\,. \tag{3}$$

Proof sketch Write the squared pairwise distance as a sum of a signal term and a noise term, $\|\mathbf{x}_i - \mathbf{x}_j\|^2 = \|\mathbf{s}_i - \mathbf{s}_j\|^2 + \|\boldsymbol{\epsilon}_i - \boldsymbol{\epsilon}_j\|^2$, where $\mathbf{s}_k$ is the signal component of point k. The noise term follows a χ^2 distribution with d_a degrees of freedom, so its mean is $2\sigma^2 d_a$ and its variance is $8\sigma^4 d_a$. By standard concentration of measure results [6], this noise term concentrates around its mean at a rate of $O(d_a^{-1/2})$, while the signal term $\|\mathbf{s}_i - \mathbf{s}_j\|$ is bounded below by some fixed cluster separation that depends only on d_i, not d_a. Putting these together, the signal-to-noise ratio in Euclidean distance behaves like $O(\sqrt{d_i/d_a}) = O(\rho^{-1/2})$, which gives the bound above. This lines up reasonably well with the empirical power law $\mathrm{ARI}(\rho) \approx \alpha \cdot \rho^{-\beta}$ that we observe in Section 5: the theoretical bound corresponds to $\beta = 0.5$, and the fact that we observe $\beta > 0.5$ for larger d_i likely reflects additional effects from how k-means estimates centroids, which we don't attempt to model here. □

4 Experimental Setup

4.1 Synthetic Data Generation

To test these ideas in a controlled way, we generate synthetic datasets by embedding d_i-dimensional Gaussian data into a d_a-dimensional ambient space via a random projection, then adding isotropic Gaussian noise:

$$\mathbf{X} = \mathbf{X}_{\mathrm{low}}\mathbf{P} + \epsilon, \quad \mathbf{X}_{\mathrm{low}} \in \mathbb{R}^{n\times d_i}, \quad \mathbf{P} \in \mathbb{R}^{d_i\times d_a}, \quad \epsilon \sim \mathcal{N}(0, \sigma^2\mathbf{I}) \tag{4}$$

where the columns of $\mathbf{P}$ are normalized. We sweep $d_i \in \{2, 3, 5, 8, 10\}$ and $\rho \in \{1, 2, 5, 10, 20, 30, 50, 75, 100\}$, giving 45 conditions in total, each repeated over 5 trials with different random seeds.

4.2 Algorithms

We test three algorithms, chosen to represent fairly different approaches to learning from data:

- **Clustering:** k-means with $k = 3$, evaluated using the Adjusted Rand Index (ARI). We treat ARI < 0.5 as a failure.
- **Classification:** a k-nearest neighbor classifier ($k = 5$), evaluated with 3-fold cross-validation accuracy.
- **Regression:** a k-nearest neighbor regressor ($k = 5$), evaluated with 3-fold cross-validation R^2.

4.3 Intrinsic Dimension Estimation

Intrinsic dimension is estimated using TwoNN [4], which compares the distances from each point to its first and second nearest neighbors. As a sanity check, we also computed MLE-based estimates on a subset of conditions and found the results were not sensitive to this choice.

4.4 Baseline Measures

To put ρ in context, we compare it against several alternative measures of "effective complexity": the ambient dimension d_a alone, the intrinsic dimension d_i alone, the participation ratio (PR), the explained variance ratio at a 50% threshold (EVR-50), and the log condition number of the covariance matrix.

5 Results

5.1 Power Law Decay and a Universal Failure Threshold

Figure 1 shows k-means ARI as a function of ρ, separately for each intrinsic dimension we tested. The pattern is strikingly consistent: ARI decreases monotonically with ρ, and the decay is well described by a power law,

$$\mathrm{ARI}(\rho, d_i) \approx \alpha \cdot \rho^{-\beta(d_i)}. \tag{5}$$

What stands out most is *where* the failure happens. If we define the phase transition as the point where the failure rate crosses 50%, this transition occurs at $\rho^* \approx 3.5$–5 for every value of d_i we tested – whether $d_i = 2$ or $d_i = 10$. In other words, the absolute values of d_a and d_i can vary enormously, but as long as their ratio crosses this threshold, k-means starts to fail.

5.2 The Decay Exponent Follows a Logarithmic Law

While the threshold ρ^* is roughly constant across d_i, the *steepness* of the decay, β, is not – it increases gradually with d_i. Fitting β for each value of d_i and then fitting a curve to those values, we find:

$$\beta(d_i) = 0.140 \cdot \ln(d_i) + 0.534 \tag{6}$$

This fits the data well – prediction error stays below 0.025 across all five values of d_i we tested (Table 1). Practically, this means that given just ρ and d_i, we can predict roughly how badly k-means will perform, using only two numbers.

Table 1 Fitted decay exponents and prediction errors.

d_i	α	β (fitted)	β (predicted)	Error
2	1.270	0.607	0.631	0.024
3	1.311	0.711	0.687	0.024
5	1.421	0.777	0.759	0.018
8	1.403	0.813	0.824	0.011
10	1.456	0.849	0.856	0.007

5.3 k-means and kNN Respond to ρ in Opposite Ways

This is, in our view, the most interesting result in the paper. Figure 2 shows what happens to k-means, kNN classification, and kNN regression as ρ increases, side by side. k-means degrades sharply, as already discussed (Spearman $r = -0.913$, $p = 2.65 \times 10^{-18}$). But kNN classification and regression do the opposite – both *improve* as ρ grows (Spearman $r = +0.608$ and $+0.600$ respectively, both $p < 10^{-5}$).

At first this seems counterintuitive – shouldn't more "noise dimensions" hurt every algorithm? But it makes sense geometrically. As ρ increases, the extra noise dimensions concentrate pairwise distances around their mean. For k-means, this is bad news: cluster centroids that were once well-separated start to look similar, and the algorithm has trouble telling clusters apart. But for kNN, this same concentration effect makes local neighborhoods more uniform, which can actually help – it reduces the algorithm's sensitivity to noisy local density variations. The same underlying geometric change produces opposite outcomes depending on whether an algorithm relies on global centroid structure or local neighborhood structure.

5.4 ρ Outperforms Existing Complexity Measures

Table 2 compares how well different measures predict k-means failure across our 45 conditions. $\log(\rho)$ comes out on top with $R^2 = 0.556$, ahead of every other individual measure we tried. Perhaps the clearest illustration of why ρ matters is that d_i on its own is essentially useless as a predictor ($R^2 = 0.032$) – it's only when you compare d_i to d_a that it becomes informative.

We also checked whether combining predictors helps. Using ordinary least squares on the same 45 conditions, combining $\log(\rho)$ with PR gives $R^2 = 0.590$; adding EVR-50 on top brings this to $R^2 = 0.605$; and using all seven predictors together gives $R^2 = 0.625$. These gains over $\log(\rho)$ alone are fairly small, which suggests $\log(\rho)$ is already capturing most of what matters – and it has the advantage of being a single, easy-to-compute number.

Table 2 Predictor comparison for k-means ARI. Higher R^2 is better.

Predictor	Spearman r	p-value	R^2
d_a (ambient dim)	-0.728	1.43×10^{-8}	0.147
d_i (intrinsic dim)	-0.501	4.57×10^{-4}	0.032
ρ (stretching ratio)	-0.542	1.20×10^{-4}	0.203
$\log(\rho)$	-0.542	1.20×10^{-4}	**0.556**
Participation Ratio	-0.856	6.46×10^{-14}	0.321
EVR-50	-0.848	2.08×10^{-13}	0.313
log(Condition Number)	-0.639	2.31×10^{-6}	0.351

5.5 Heavy-Tailed Noise Makes Things Worse, Earlier

So far we've assumed Gaussian noise in the ambient dimensions. What happens under heavier-tailed noise? We repeated the main experiment using t-distributed noise with 2 degrees of freedom – a substantially heavier-tailed distribution – and found two things. First, even at $\rho = 1$, ARI drops by up to 36% compared to the Gaussian case: heavy tails hurt performance even before any "stretching" effect kicks in. Second, the failure threshold itself shifts: ρ^* drops to below 2, compared to roughly 3.5–5 under Gaussian noise. This suggests that the stretching ratio and the noise distribution's tail behavior interact multiplicatively rather than independently, which is broadly consistent with prior findings on distance concentration under non-Gaussian distributions [7].

5.6 Does This Hold on Real Data?

Synthetic experiments are useful because we can control exactly what's happening, but the real test is whether ρ says anything useful about real datasets. We checked this on three datasets.

Wine (UCI).

The Wine dataset has 178 samples across 3 cultivar classes, described by 13 continuous features. To vary ρ in a controlled way, we appended blocks of pure Gaussian noise dimensions, leaving the original cluster structure intact. The results (Table 3) line up well with the synthetic findings: ARI drops from 0.897 at $\rho \approx 1.65$ to 0.208 at $\rho \approx 3.36$ – right around the $\rho^* \approx 3.5$–5 range we identified earlier.

MNIST.

For MNIST, we used PCA to reduce the original 784 dimensions down to several smaller values ($d_a \in \{10, 20, 50, 100, 200, 784\}$), estimating d_i via TwoNN on a 500-sample subsample at each setting. Here the picture is different: ARI stays roughly constant across all values of ρ we tried (Table 3). Our interpretation is that for MNIST, k-means is already limited by the non-convex shape of the digit clusters themselves, not by noise-driven distance dilution – so the stretching-ratio effect, while presumably still present, is masked by this other limitation.

Olivetti Faces.

The Olivetti dataset (400 images, 40 individuals, 4096 features originally) shows a similar pattern to MNIST: performance appears to be bounded by overlap between classes in face-space, rather than by ambient dimensionality.

Table 3 Real-world validation results. d_a and d_i are estimated values; † marks conditions where noise dimensions were appended.

Dataset	Condition	d_a	d_i (est.)	ρ (est.)	ARI
Wine†	Original	13	7.9	1.65	0.897
Wine†	+10 noise dims	23	8.1	2.84	0.621
Wine†	+20 noise dims	33	9.8	3.36	0.208
Wine†	+50 noise dims	63	10.2	6.18	0.041
MNIST (PCA)	$d_a = 784$	784	38.2	20.5	0.361
MNIST (PCA)	$d_a = 100$	100	35.1	2.85	0.374
MNIST (PCA)	$d_a = 20$	20	13.4	1.49	0.353
Olivetti	Full ($d_a = 4096$)	4096	41.3	99.2	0.312
Olivetti	PCA ($d_a = 100$)	100	38.7	2.58	0.318

So Wine reproduces the synthetic threshold quite closely, while the two image datasets reveal an important boundary condition: ρ predicts failure well specifically in the regime where distance dilution is the main bottleneck. When something else – like non-convex cluster shapes – is already the limiting factor, ρ has less to say.

6 Discussion

What does this mean in practice?

The most direct takeaway is a simple diagnostic: before running k-means (or similar centroid-based clustering), estimate ρ using TwoNN. If $\rho > 5$, the results are likely to be poor regardless of how much you tune k or the algorithm's other parameters – some form of dimensionality reduction is probably necessary first. For kNN-based methods, on the other hand, a high ρ is not a red flag, and may even help slightly.

Why does ρ cut both ways?

The opposite responses of k-means and kNN to increasing ρ have a fairly intuitive geometric story. As ρ grows, most of the ambient dimensions are contributing noise rather than signal, and this noise concentrates pairwise distances – everything starts to look roughly equidistant from everything else. For k-means, which relies on clusters having distinguishable centroids, this is destructive. For kNN, which relies on local neighborhood structure, this same concentration can smooth out noisy local variations and slightly improve performance.

Limitations.

Everything here is empirical – we don't have a full theoretical derivation of either the power law form or the logarithmic scaling of β, only the partial argument in Proposition 1. Our real-data validation relied on injecting noise dimensions to vary ρ, which is a reasonable way to isolate the effect we care about but may not perfectly mirror how ρ varies "naturally" across real datasets. It's also worth noting that intrinsic dimension estimation becomes harder and less reliable for deep learning representations, so it's not yet clear how well these results would transfer to that setting.

Where this could go next.

A few directions seem natural: working out a tighter theoretical justification for the scaling laws we observe, testing whether the same ρ-dependent patterns hold for other algorithms (SVMs, Gaussian processes, spectral clustering), seeing how this plays out for learned representations rather than raw features, and potentially building ρ-aware preprocessing tools that automatically flag datasets likely to cause trouble for centroid-based methods.

7 Conclusion

We introduced the stretching ratio $\rho = d_a/d_i$ as a way of predicting how machine learning algorithms behave on high-dimensional data. Across 45 controlled experiments, k-means clustering performance follows a power law in ρ with a failure threshold around $\rho^* \approx 3.5$–5 that holds across different intrinsic dimensions, and the rate of decay follows a simple logarithmic relationship with d_i. Perhaps most notably, kNN-based classification and regression respond to ρ in the opposite direction – improving rather than degrading – which suggests that what's really going on is a geometric trade-off rather than a single, uniform "curse." We also found that ρ is a better predictor of clustering failure than several existing complexity measures, while remaining simple enough to compute in practice. We hope this gives practitioners a useful, lightweight diagnostic, and gives researchers a concrete pattern to explain theoretically.

Data Availability

All code used to generate the synthetic datasets, run the experiments, and produce the figures in this paper, along with the raw experimental results, is publicly available at https://github.com/daykodotexe/ambient-intrinsic-gap.

Declarations

Conflict of interest: The author declares no conflict of interest.
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Figures

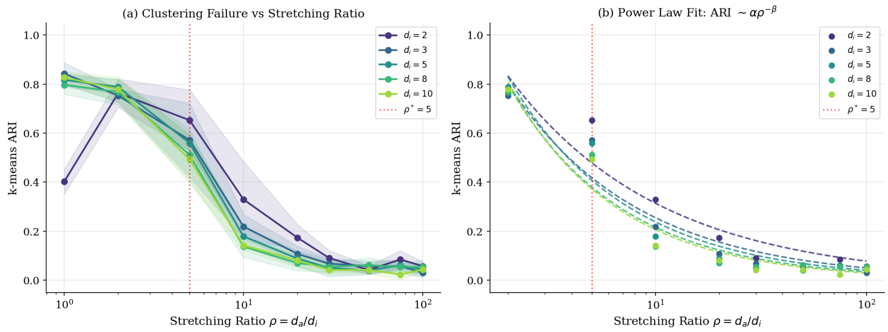

Fig. 1 (a) k-means ARI decreases monotonically with stretching ratio ρ. Shaded regions show ± 1 standard deviation across trials. The red dotted line marks $\rho^* = 5$. (b) Power law fits ARI $\sim \alpha\rho^{-\beta}$ across all intrinsic dimensions.

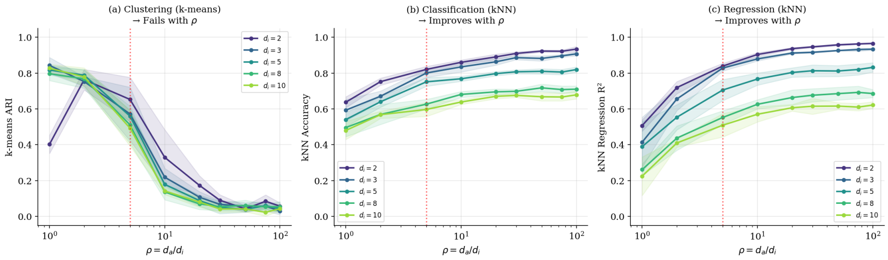

Fig. 2 Algorithm-dependent effects of ρ. k-means degrades (left) while kNN classification (center) and regression (right) improve. This divergence reveals a geometric tradeoff rather than a simple curse.

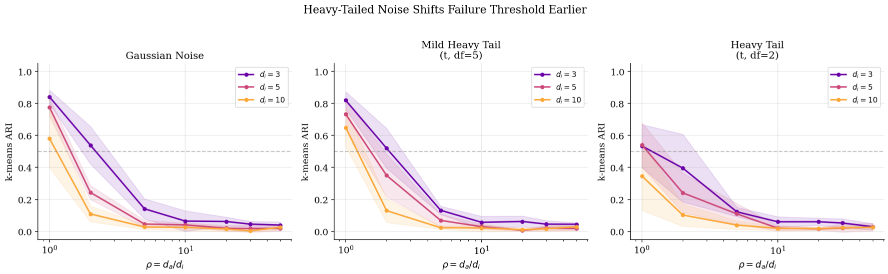

Fig. 3 Heavy-tailed noise shifts the failure threshold earlier. Under t-distribution noise with df=2, failure occurs at $\rho^* < 2$ compared to $\rho^* \approx 3$–5 under Gaussian noise.

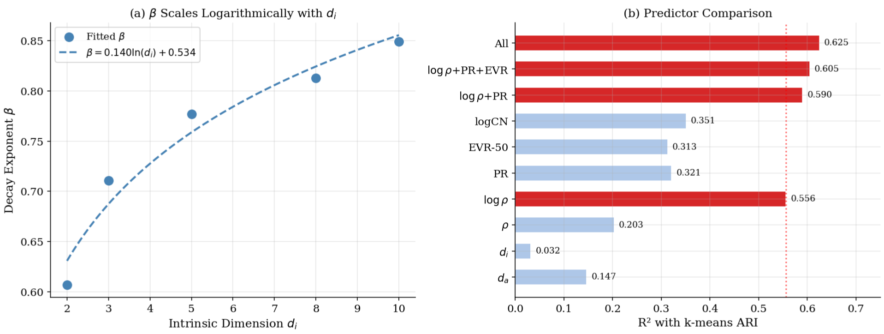

Fig. 4 (a) Decay exponent β scales logarithmically with d_i. (b) $\log(\rho)$ achieves the highest R^2 among individual predictors, outperforming participation ratio and explained variance ratio.